

PROBLEM 1: MOVING AVERAGES

Consider $x_0, x_1, \dots$ and two running averages defined as, for $0 < \beta < 1$ and $\gamma > 0$.

$$\begin{aligned} y_0 &= 0 \\ y_{t+1} &= \beta y_t + (1 - \beta)x_t \end{aligned}$$

$$\begin{aligned} z_0 &= 0 \\ z_{t+1} &= \beta z_t + \gamma x_t \end{aligned}$$

Part (a). Suppose x_t i.i.d. from distribution with mean vector $\bar{x}$ Give values for

$$\bar{y} = \lim_{t \rightarrow \infty} \mathbb{E} y_t$$

and

$$\bar{z} = \lim_{t \rightarrow \infty} \mathbb{E} z_t$$

as functions of β, γ , and $\bar{x}$.

Proof.

$$\begin{aligned} \mathbb{E} y_2 &= \beta \mathbb{E} y_1 + (1 - \beta)\bar{x} \\ &= \beta(\beta \mathbb{E} y_0 + (1 - \beta)\bar{x}) + (1 - \beta)\bar{x} \\ &= \beta^2 \mathbb{E} y_0 + \beta(1 - \beta)\bar{x} + (1 - \beta)\bar{x} \\ &= (1 - \beta)\bar{x} \sum_{i=0}^1 \beta^i \\ \implies \mathbb{E} y_t &= (1 - \beta)\bar{x} \sum_{i=0}^{t-1} \beta^i \\ \implies \bar{y} = \lim_{t \rightarrow \infty} \mathbb{E} y_t &= (1 - \beta)\bar{x} \sum_{i=0}^{\infty} \beta^i \\ &= \bar{x} \end{aligned}$$

For z_t , we have

$$\begin{aligned} \mathbb{E} z_3 &= \beta \mathbb{E} z_2 + \gamma \bar{x} \\ &= \beta(\beta \mathbb{E} z_1 + \gamma \bar{x}) + \gamma \bar{x} \\ &= \beta^2 \mathbb{E} z_1 + \beta \gamma \bar{x} + \gamma \bar{x} \\ &= \beta^2(\beta \mathbb{E} z_0 + \gamma \bar{x}) + \beta \gamma \bar{x} + \gamma \bar{x} \\ &= \beta^3 \mathbb{E} z_0 + \gamma \bar{x} \sum_{i=0}^2 \beta^i \\ \implies \mathbb{E} z_t &= \gamma \bar{x} \sum_{i=0}^{t-1} \beta^i \\ \implies \bar{z} = \lim_{t \rightarrow \infty} \mathbb{E} z_t &= \gamma \bar{x} \sum_{i=0}^{\infty} \beta^i \\ &= \frac{\gamma}{1 - \beta} \bar{x} \end{aligned}$$

□

Part (b). Express z_t as a function of y_t, β , and γ .

Proof. From part (a) we have:

$$z_t = \gamma \sum_{i=0}^{t-1} \beta^i x_{t-1-i}$$

$$y_t = (1 - \beta) \sum_{i=0}^{t-1} \beta^i x_{t-1-i}$$

This implies that

$$z_t = \frac{\gamma}{1 - \beta} y_t$$

□

PROBLEM 2: VARIANCE OF AN EXPONENTIAL MOVING AVERAGE

Consider a running average for computing $\hat{\mu}_1, \dots, \hat{\mu}_t$ from $x_1, \dots, x_t$

$$\hat{\mu}_0 = 0$$

$$\hat{\mu}_t = \left(1 - \frac{1}{N}\right) \hat{\mu}_{t-1} + \frac{1}{N} x_t$$

Part (a). Suppose x_t are i.i.d. with variance σ_x^2 . Find

$$\sigma_{\hat{\mu}, \infty}^2 = \lim_{t \rightarrow \infty} \text{Var } \hat{\mu}_t$$

Proof. Since $x_t \perp \hat{\mu}_{t-1}$, we have

$$\begin{aligned} \text{Var } \hat{\mu}_t &= \left(1 - \frac{1}{N}\right)^2 \text{Var } (\hat{\mu}_{t-1}) + \frac{1}{N^2} \text{Var } x_t \\ &= \left(1 - \frac{1}{N}\right)^2 \text{Var } (\hat{\mu}_{t-1}) + \frac{\sigma_x^2}{N^2} \end{aligned}$$

Taking limits on both sides, we have

$$\begin{aligned} \sigma_{\hat{\mu}, \infty}^2 &= \left(1 - \frac{1}{N}\right)^2 \sigma_{\hat{\mu}, \infty}^2 + \frac{\sigma_x^2}{N^2} \\ \left(1 - \left(1 - \frac{1}{N}\right)^2\right) \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \left(1 - 1 + \frac{1}{N}\right) \left(1 + 1 - \frac{1}{N}\right) \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \frac{2N - 1}{N^2} \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{2N - 1} \end{aligned}$$

□

Part (b). Compare $\sigma_{\hat{\mu}, \infty}^2$ to the variance of

$$\hat{\mu} = \frac{1}{N} \sum_{t=1}^N x_t$$

Proof.

$$\begin{aligned} \text{Var } \hat{\mu} &= \frac{1}{N^2} \sum_{t=1}^N \text{Var } x_t \\ &= \frac{\sigma_x^2}{N} \end{aligned}$$

□

PROBLEM 3. REFORMULATING MOMENTUM AS AN EXPONENTIAL MOVING AVERAGE

Consider the following update equation.

$$\begin{aligned} y_0 &= 0 \\ y_t &= \left(1 - \frac{1}{N}\right) y_{t-1} + x_t \end{aligned}$$

Part (a). Assume y_t converges to a limit. If the input sequence is constant $x_t = c$ what is the limit?

Proof. We have the following

$$\begin{aligned} y &= \left(1 - \frac{1}{N}\right) y + c \\ y &= Nc \end{aligned}$$

□

Part (b). y_t is an exponential moving average of what quantity?

Proof. y_t is an exponential moving average of Nx_t

□

Part (c). Express y_t as a function of μ_t where μ_t is the exponential moving average of x_t with parameter $\frac{1}{N}$.

Proof. Recall that we have:

$$y_t = N \left(1 - \frac{1}{N}\right) \sum_{i=0}^{t-1} \frac{1}{N^i} x_{t-1-i}$$

On the other hand, we have

$$\mu_t = \left(1 - \frac{1}{N}\right) \sum_{i=0}^{t-1} \frac{1}{N^i} x_{t-1-i}$$

This gives us that

$$y_t = N\mu_t$$

□

PROBLEM 4. BIAS CORRECTION

Consider the following update equation for computing $y_1, \dots, y_t$ from $x_1, \dots, x_t$.

$$y_t = \left(1 - \frac{1}{t \wedge N}\right) y_{t-1} + \frac{1}{t \wedge N} x_t$$

If $x_t = c$ give a closed form solution for y_t .

Proof. Let $t < N$. We have:

$$\begin{aligned} y_t &= \left(1 - \frac{1}{t}\right) y_{t-1} + \frac{1}{t} c \\ &= \left(1 - \frac{1}{t}\right) \left(\left(1 - \frac{1}{t-1}\right) y_{t-2} + \frac{1}{t-1} c \right) + \frac{1}{t} c \\ &= \left(1 - \frac{1}{t}\right) \left(1 - \frac{1}{t-1}\right) y_{t-2} + \left(1 - \frac{1}{t}\right) \frac{1}{t-1} c + \frac{1}{t} c \\ &= c \prod_{i=2}^t \left(1 - \frac{1}{i}\right) + c \sum_{i=1}^t \frac{1}{i} \prod_{j=i+1}^t \left(1 - \frac{1}{j}\right) \end{aligned}$$

Alternatively,

$$y_1 = c$$

and for $t > 1$, we have that y_{t+1} is a convex combination of y_t and c . Since $y_1 = c$, we have that $y_t = c$ for all t .

□